ME 5309 – Mid-Semester Examination: ANSWER SHEET

Roll No.: ME25M032 **Name:** Pranav Deshpande

Questions attempted: Q1, Q2, Q3

Question 1

(a) (i) $f_x = f / p = 16\text{ mm} / 5\ \mu\text{m} =$ **3200 px**.

(ii) $W_s = 1280 \times 5\ \mu\text{m} = 6.4$ mm. FOV $= W_s \cdot Z/f = 6.4 \times 400/16 =$ **160 mm**.

Per pixel $= 160/1280 \approx$ **0.13 mm/px**.

(b) Resolution of the ADC $= V_{range}/2^n = 5\text{ V} / 4096 = 1.22\times10^{-3}$ V = **1.22 mV**.

So the smallest change the system can resolve is 1.22 mV.

(c) $F = e^{AT}$, $G = A^{-1}(F - I)B$ for a ZOH input.

$F = e^{-0.2} =$ **0.82**; $G = (0.82 - 1)/(-2) =$ **0.09**. So $x_{k+1} = 0.82\ x_k + 0.09\ u_k$.

Question 2

(a) States $x_1 = \theta$, $x_2 = \dot{\theta}$. $\dot{x}_1 = x_2$; $\dot{x}_2 = (1/J)[\tau - b\ x_2 - mg\ell \sin x_1]$.

Equilibria ($\tau = \tau_0$, $\dot{x} = 0$): $x_2 = 0$ and $mg\ell \sin \theta_e = \tau_0$. Since $|\sin\theta| \le 1$, an equilibrum exists iff $|\tau_0| \le mg\ell = 2.4525$ N·m.

(b) $A = [0,\ 1;\ -(mg\ell/J)\cos\theta_e,\ -b/J] = [0,\ 1;\ -122.6 \times 0.5,\ -2.5] =$ **[0, 1; −61, −2.5]**.

$B = [0;\ 1/J] =$ **[0; 50]**.

(c) At rest $\dot{\theta} = \ddot{\theta} = 0$, so the PD torque must balance gravity: $K_p(\theta_d - \theta_{ss}) = mg\ell \sin\theta_{ss}$.

If $\theta_{ss} = \theta_d = 90°$ the left side is 0 but the right side is $mg\ell \sin 90° = 2.4525 \neq 0$: contradiction. So the joint cannot rest at θ_d; the graviy torque needs a non-zero error to produce the holding torque.

Solve $50(\pi/2 - \theta_{ss}) = 2.4525 \sin\theta_{ss}$ numerically: $\theta_{ss} \approx 1.5218$ rad = 87.19°. Error $\approx 90° - 87.19° =$ **2.81°** (joint sags below θ_d).

(d) With compensaton, at rest: $K_p(\theta_d - \theta_{ss}) + mg\ell \sin\theta_{ss} = mg\ell \sin\theta_{ss} \Rightarrow \theta_{ss} = \theta_d$: **steady-state error = 0**.

Closed loop: $J\ddot{\theta} + (b + K_d)\dot{\theta} + K_p(\theta - \theta_d) = 0$; graviy cancels. Without compensaton the linearised stiffness is $K_p + mg\ell\cos\theta_d$, and at $\theta_d = 90° \cos\theta_d = 0$, so the stiffness is K_p either way. **Poles unchanged** (specific to 90°; at other angles they would shift).

Question 3

(a) (1) Taking the difference of two readings amplifies noice.

(2) The high frequency noise in the encoder signal gets magnified when we subtract successive readings and divide by a small T.

(b) $\omega_c = 2\pi(20) = 125.7$ rad/s; Tustin with T = 1 ms, $a = \omega_c T/2 = 0.063$.

$y_k =$ **$0.059\ (u_k + u_{k-1}) + 0.88\ y_{k-1}$**.

DC gain: $2(0.059)/(1 - 0.88) = 0.118/0.12 \approx 0.98 \approx 1$ ✓ (rounding).

(c) $A - LC = [-\ell_1,\ 1;\ -\ell_2,\ -2]$. $\det(sI - A + LC) = (s + \ell_1)(s + 2) + \ell_2 = s^2 + (2 + \ell_1)s + (2\ell_1 + \ell_2)$.

Desired: $(s + 10)^2 = s^2 + 20s + 100$. Match: $2 + \ell_1 = 20 \Rightarrow \ell_1 = 18$; $2\ell_1 + \ell_2 = 100 \Rightarrow \ell_2 = 64$. **$L = [18, 64]^T$**.

(d) Advantage of the obsever: it uses the plant model and the applied input u, so its velocity estimate has (almost) no phase lag; the 20 Hz filter on differenced data always delays by about

$1/(2\pi f_c) \approx 8$ ms. It can also reconstruct states that cannot be obtained by differencing (e.g. disturbance, when augmented).

Filter is safer when the model is uncertain or changes (unknown payload, friction, contact, actuator saturation): the observer then gives a biased or poorly-behaved estimate while the model-free filter still follows the measured data.